

B03 Gate A Protected Review - Fleet Maintenance

Protected diagnostic review - preserve fresh A/B/C evidence

Protected review - saved genuine Gate attempt required

Use this only after the matching genuine Gate attempt is saved. Diagnose, repair, close the review, then produce changed evidence. Later Gates remain state-machine protected.

Objective source controls

vehicles 20; technicians 8; work orders 48; part lines 96; labor logs 72; certifications 25

Metric	Protected control
Base service cost	121350.00
Parts cost	64125.00
Labor cost	90405.00
Closed work orders	32
Naive work-order x parts x labor rows	144
Naive repeated base-service total	367200.00
Naive repeated parts total	97000.00
Naive repeated labor total	181755.00

Why a plausible model can still be wrong

- work_orders, work_order_parts and labor_logs are separate grains. A raw three-way join multiplies all three measures.
- vehicle_certifications is a real membership bridge/factless-fact candidate; five vehicles have more than one certification.
- Do not repair fan-out with DISTINCT after the join. Fix the model path and aggregate at each fact grain.

Reasoning and evidence rubric

- State each business process and exact one-row grain before assigning fact/dimension/bridge roles.
- Prove dimension/business-key uniqueness and state foreign-key/cardinality expectations.
- Separate facts at different grains. Predict and test fan-out before any cross-fact aggregation.
- Use bridges only for legitimate membership; never as a broom for duplicate dimension defects.
- Keep role-playing dates explicit and name the question each date role answers.
- Reconcile raw counts/totals to modeled outputs. Unexplained differences remain visible blockers.
- Provide an implementation-ready handoff plus a rejected alternative and why it fails.

Critical failures

- Mixed grain or hidden fan-out.
- Arbitrary many-to-many/DISTINCT repair.
- Target or transaction measures repeated at a lower/higher incompatible grain.
- Failed reconciliation hidden or explained only by “the visual looks right.”
- Cannot reproduce/defend the model contract.

Repair and changed retry route

- Save the original failed/developing Gate attempt.
- Open only this matching protected review after the saved attempt exists.
- Repair the named weak competency using the targeted remediation cards.
- Close this review and produce a changed micro-proof in a different small scenario.
- Save the changed evidence. Gate B/C still require the existing explicit fresh full-retest assignment when applicable.

Next route

PASS: B03 ends and course continues to B04. FAIL/developing: repair; Gate B stays sealed.